

Supplemental Material for *Optimization-based hologram design for fine optical tweezer arrays and extension of super-resolution criteria*

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Vector Debye formula for focal pattern under high-NA focusing conditions.

To accurately calculate the diffraction integral, the vector nature of light must be considered to obtain focal patterns that are deemed correct within wave optics [36, 37]. The vector nature of light in the focusing phenomenon is schematically illustrated in Fig. 1. An objective lens modifies the incident light by attaching a spherical wavefront, thereby altering the propagation direction of light towards the center of the spherical wavefront, known as the geometric focus [Fig. 1(a)]. As the propagation direction changes, the vector amplitude of the incident light is reduced, inducing a polarization component in the direction of light propagation [Fig. 1(b)]. This results in an inhomogeneous modulation of the vector light amplitude due to the edge birefringence of the lens [Fig. 1(c)] [47]. These characteristics are specific to focusing by a high-NA objective lens, and this phenomenon is referred to as depolarization.

The polarization vectors of light at the entrance pupil and image space of an objective lens are expressed in Cartesian coordinates, where the z -axis and origin correspond to the optical axis and geometrical focus, respectively [Fig. 1(a)]. Although the coordinates of the entrance pupil and those in the image space are distinct, polarization vectors can be described in a common Cartesian coordinate system. In general, various coordinate systems are introduced to describe the positions on the entrance pupil and in the image space. The former is expressed in polar coordinates, (r, φ) , whereas the latter uses cylindrical coordinates (ρ, ϕ, z) . The cylindri-

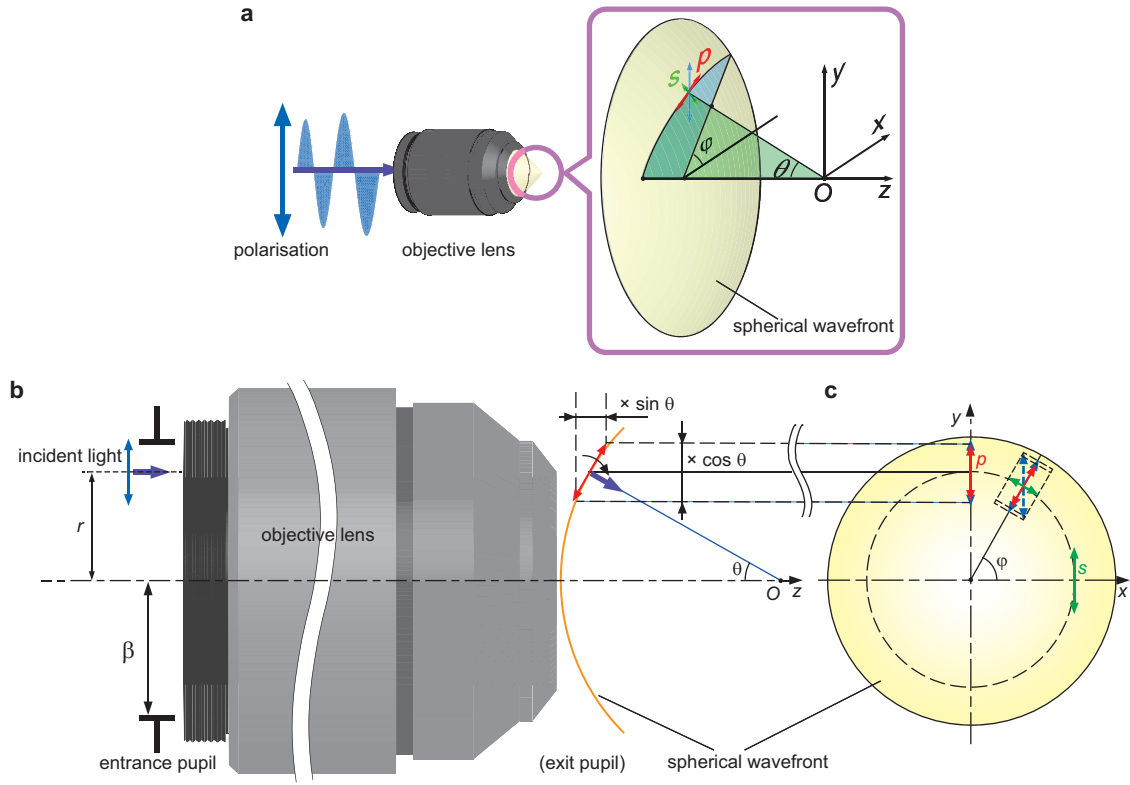


Fig. 1S Vector nature of light amplitude in focusing. (a) Focusing of linearly polarized light through an objective lens. Vector amplitude of incident flat-wavefront light is mapped to that on spherical wavefront at the exit pupil of the objective lens. A position on the spherical wavefront is identified by the cone and directional angles, θ and φ , respectively. (b) Cross-sectional view of (a) on a $z - \varphi$ plane. φ -directional component of the vector amplitude of incident light (blue arrows) is mapped to that (red arrows) on spherical wavefront by the objective lens. (c) Front view of (a). The objective lens tilts the p -component (radial component: red arrows) of a vector light amplitude but does nothing to the s -component (azimuthal component: green arrows). This causes inhomogeneous change in the light polarization, because the decomposition into the p - and s -components depends on φ for linearly-polarized incident light.

cal and Cartesian coordinates in the image space are compatible through a coordinate transformation. Note that the azimuth ($= \varphi$) at the entrance pupil represents the directional angle of the spherical wavefront at the exit pupil [Fig. 1(a)] because an ideal objective lens does not alter the azimuthal direction. The vector Debye formula defines an integral transformation of the

vector field of light amplitude from (r, φ) to (ρ, ϕ, z) , as determined by the action of a high-NA objective lens.

In the context of this study, the incident light is assumed to be monochromatic, with a wavelength λ ($= 2\pi/k$). We assumed that the incident light has a spatially homogeneous polarization and electric field amplitude distribution of the form $e^{-il\varphi} E_{\text{in}}(r)$. The focal patterns of various light beams can be derived, in principle, from a combination of these distributions. The polarization vector of the incident light is expressed as a unit vector $\hat{e}_{\text{in}}$, where the components are a pair of complex numbers, α and β ($|\alpha|^2 + |\beta|^2 = 1$). Specifically, $\hat{e}_{\text{in}} = (\alpha, \beta) = \alpha\hat{x} + \beta\hat{y}$, where $\hat{x}$ and $\hat{y}$ represent unit vectors in the x and y directions, respectively. In particular, $(\alpha, \beta) = (1/\sqrt{2}, \mp i/\sqrt{2})$ describes the left or right circular polarization.

When the incident light is expressed by the vector amplitude: $e^{-il\varphi} E_{\text{in}}(r)\hat{e}_{\text{in}}$ at the entrance pupil of the objective lens, electric field amplitude $\mathbf{E}$ at the observation point (ρ, ϕ, z) in the image space is expressed as follows [39, 40]:

$$\begin{aligned} \mathbf{E}(\rho, \phi, z) = & \pi i^l k^2 \int_0^{\gamma_{\text{max}}} \sin \theta d\theta \mathcal{E}(\theta) \exp(ikz \cos \theta) e^{-il\phi} \\ & \times \left\{ (1 + \cos \theta) J_l(k\rho \sin \theta) \hat{e}_{\text{in}} \right. \\ & - \sqrt{2}i \sin \theta [\langle \hat{e}_+, \hat{e}_{\text{in}} \rangle J_{l+1}(k\rho \sin \theta) e^{-i\phi} \\ & \quad - \langle \hat{e}_-, \hat{e}_{\text{in}} \rangle J_{l-1}(k\rho \sin \theta) e^{i\phi}] \hat{e}_z \\ & + (1 - \cos \theta) [\langle \hat{e}_+, \hat{e}_{\text{in}} \rangle J_{l+2}(k\rho \sin \theta) e^{-2i\phi} \hat{e}_- \\ & \quad \left. + \langle \hat{e}_-, \hat{e}_{\text{in}} \rangle J_{l-2}(k\rho \sin \theta) e^{2i\phi} \hat{e}_+] \right\}, \quad (\text{S1}) \end{aligned}$$

where $\mathcal{E}(\theta)$ denotes the θ distribution of the incident-light amplitude [48] and γ_{max} denotes the maximum focusing angle [$= \sin^{-1}(\text{NA})$]. Moreover, the inner product of the complex vectors is expressed as $\langle \mathbf{a}, \mathbf{b} \rangle (= \sum_i a_i^* b_i)$ in Eq. (S1).

The relationship between $E_{\text{in}}(r)$ and $\mathcal{E}(\theta)$ is determined by the optical design of the focusing lens. For example, $E_{\text{in}}(\beta \sin \theta / \sin \gamma_{\text{max}}) = \mathcal{E}(\theta)$ holds for objective lenses that satisfy the sine condition (here, β indicates the radius of the entrance pupil). Based on equation (S1), we can confirm that the circular-polarization flat-wavefront light (i.e., circularly-polarized light of $l = 0$) presents a focal pattern identical to the mean of those of linear-polarization (e.g., x - and

y -polarizations) flat-wavefront light.

Airy-disk radius as function of NA.

Since the super-resolution criterion proposed in the main text is established by selecting the radius of an Airy disk as a unit of the length, it is necessary to know the actual value of the Airy-disk radius for determining whether an observed light-spot array exceeds the super-resolution criterion or not under a particular optical condition. From an opposite point of view, it is possible to apply the proposed super-resolution criterion to another optical condition by replacing the Airy-disk radius with that derived properly according to the condition under consideration.

An Airy disk was originally defined as the diffraction pattern of a circular aperture in the far-field region of light propagation and is derived from the scalar diffraction formula, which is proportional to $[J_1(k\rho\text{NA})/(k\rho\text{NA})]^2$ on the focal plane. The distance from the center of the image to the innermost dark ring is often referred to as the Airy disk radius, which is related to the conventional super-resolution criterion.

Equation (S1) reproduces a focal pattern that is similar to that of a conventional Airy disk. However, a finite value was observed in the innermost dark ring [Fig. 2(a)]. Therefore, the Airy-

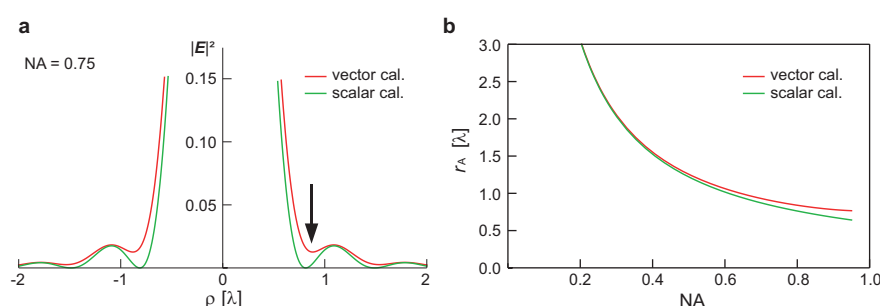


Fig. 2S Comparison between scalar and vector calculations. (a) Cross-sectional profiles of focal patterns calculated by scalar diffraction formula (green line) and vector Debye integral (red line). Both of the profiles were calculated under the condition of $\text{NA} = 0.75$. (b) Airy-disk radii as functions of NA. Results of scalar diffraction formula and vector Debye integral are plotted in green and red lines, respectively. The scale of the length is selected as the wavelength of light in both (a) and (b).

disk radius is redefined as the distance from the optical axis to the innermost local minimum in the high-NA region. The redefined Airy-disk radius is marginally larger than the conventional one, which is given by $0.61/\text{NA}$ in units of wavelength. The conventional and high-NA Airy-disk radii are depicted as functions of NA in Fig. 2(b) to illustrate the difference between them. This difference is almost negligible for smaller NA values but becomes notable for $\text{NA} \gtrsim 0.7$.

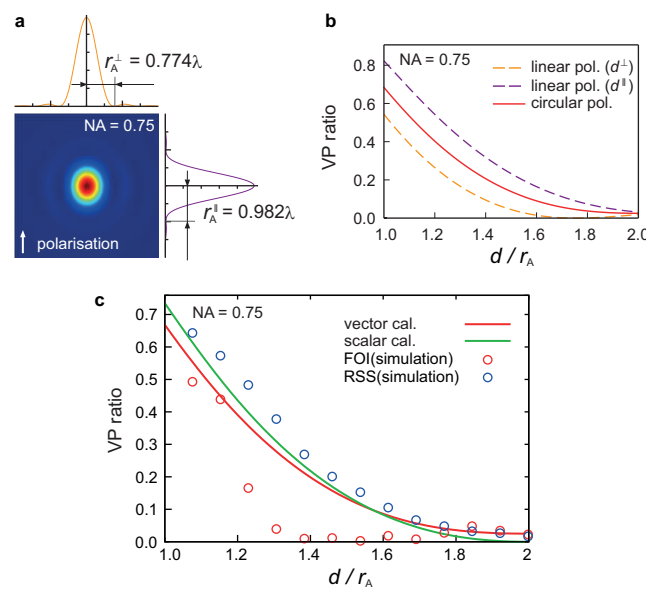


Fig. 3S Effects of polarization under condition of $\text{NA} = 0.75$. (a) Focal pattern and corresponding cross-sectional profiles of linear-polarization light. (b) Super-resolution criteria for circular-polarization (solid red line) and y linear-polarization (dashed lines) light. In the case of the linear-polarization light, the super-resolution criteria acts differently according to the light-polarization direction (dashed purple line) and the perpendicular direction (dashed orange line). (c) Comparison of numerical spot-array patterns with the super-resolution criteria. Open circles present spot spacing and VP ratio of the spot-array patterns obtained by applying the numerical hologram reproduction procedure (FFT) to holographic phase patterns designed by FOI (red open circles) and RSS (blue open circles) methods. Red and green solid curves display the super-resolution criteria calculated by the vector and scalar diffraction formulae, respectively.

Super-resolution criteria in high-NA regions.

Linear-polarization flat-wavefront light produces a focal pattern as shown in Fig. 3(a) under high-NA focusing conditions. Consequently, the Airy-disk radius has different values in the horizontal and vertical directions. Therefore, different length scales must be introduced to the horizontal and vertical spacings in a light spot array of linear-polarization light for estimation of super-resolution according to the proposed criteria in the main text.

Equation (S1) indicates that a focal pattern with circular-polarization flat-wavefront light is determined as the sum of the horizontal and vertical components. This also holds true, not strictly but effectively, up to the leading three digits of the Airy-disk radius. According to this observation, we can expect that the relation between super-resolution criteria calculated for linearly and circularly polarized light is useful for evaluating super-resolution. In fact, Fig. 3(b) reveals that the super-resolution criterion for circularly polarized light can be approximated as the mean of those for the horizontal and vertical linear-polarization lights, where the Airy-disk radius of the circularly polarized light is used commonly as the scale of length. Thus, the super-resolution criterion for circularly polarized light is expected to be applicable to the evaluation of the mean spot spacing in a square light-spot array of linearly polarized light.

To evaluate the super-resolution of a given light spot array, it is reasonable to calculate a criterion with an appropriate selection of the light propagation conditions under which the spot-array pattern was generated. As hologram design is performed by applying the FFT, specifically through Fraunhofer diffraction calculation, the super-resolution criterion for numerically generated light-spot patterns can be assessed using the scalar diffraction formula. On the other hand, the super-resolution for an experimentally observed pattern cannot be determined from the criterion based on the scalar diffraction formula, because the approximate condition of the formula is not satisfied at a $NA = 0.75$. As discussed in the main text, the VP ratio of the experimentally observed patterns exceeds the super-resolution criterion based on the vector Debye formula, indicating that super-resolution patterns were obtained using the FOI-cost method.

Nevertheless, the super-resolution criterion based on the scalar diffraction formula is expected to provide insights into estimating the potential effectiveness of the method. Figure 3(c) suggests that the FOI-cost method can potentially produce super-resolution light-spot arrays

based on the super-resolution criterion (solid green line) calculated using the scalar diffraction formula.